REFLEX: Reflexive Equilibrium Fixed-point Learning for Endogenous Financial Markets

Analytic Stability Boundaries for Performative Market Making in OTC Corporate Bond Markets

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Abstract

A machine-learned dealer reshapes the data it is trained on: tighter quotes attract more informed, adversely selecting flow, and retraining on the induced flow closes a feedback loop. This is performative prediction, whose classical stability condition $\varepsilon < \gamma/\beta$ is stated in terms of abstract constants of an unspecified loss. We present **REFLEX**, a framework that pins all three constants to the microstructure primitives of a structural over-the-counter (OTC) corporate-bond market-making model, making the stability of the retraining loop computable *a priori*, and extends the boundary in closed form to performative-gradient correction, N competing dealers ($\varepsilon < \gamma/(N_{\text{eff}}\beta)$), finite-sample robust certificates, 128-bond factor universes, and lazy deployment schedules ($\mu(K) = -m + c^K(1+m)$). Every prediction is then verified against a learned simulator loop under a common-random-number probe protocol: measured moduli track the closed form within 8% in the contracting regime; the measured stability crossing is reconciled by an independently triangulated liquidity-inflation correction; competition amplifies instability by the predicted effective dealer count; and a *structurally anchored* performative-gradient loop converges to the realized performative optimum in a regime where blind retraining collapses into the echo chamber – and where a free-form learned correction fails. Evaluated on 36 years of real market data, the boundary yields a daily fragility index whose stability headroom collapses ~4.4× from calm to crisis. All 66 load-bearing identities are machine-checked by numerical proof certificates, with formal skeletons in Lean 4.

CCS Concepts

• **Computing methodologies → Machine learning**; • **Applied computing → Economics**; • **Theory of computation →** *Market equilibria.*

Keywords

performative prediction, market making, OTC corporate bonds, repeated retraining, stability, systemic risk, optimal transport, machine-checked proofs

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ICAIF '26, Milan, Italy
2026. ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

1.1 Problem Definition

Corporate bonds trade over the counter: a client requests a quote, a dealer posts a half-spread h, and the dealer's inventory absorbs the flow. Quoting on these desks is increasingly algorithmic, and quoting policies are increasingly *fit* to historical flow. But the flow is not exogenous. A tighter quote wins more volume *and* summons more informed ("toxic") flow that picks the dealer off before adverse price moves; a wider quote starves both channels. The policy ϕ therefore induces the distribution $D(\phi)$ it will next be trained on. Repeated retraining – fit on the induced flow, redeploy, refit – is exactly *repeated risk minimization* (RRM) in the sense of performative prediction [22]: it converges to a performatively stable point if and only if $\varepsilon < \gamma/\beta$, where ε bounds the sensitivity of the distribution map, and γ, β are the strong convexity and smoothness of the loss.

The theorem is sharp, but its constants are Lipschitz *assumptions.* A desk cannot evaluate ε, β, or γ before deploying, so it cannot know ex ante whether its own retraining cadence is stable – and a regulator watching many dealers retrain against one shared pool of informed flow cannot know whether *competition among learners* is manufacturing systemic fragility. Both questions are quantitative, and both need the constants, not the abstraction. This places the problem squarely in the AI-governance and systemic-risk scope of AI in finance: the object we compute is a market-level stability margin for algorithmic liquidity provision.

1.2 Current Approaches

Two mature literatures approach the loop from opposite ends. The performative-prediction literature has sharpened the loop itself: convergence of RRM and its stochastic variants [19, 22], gradient methods that estimate the distribution response $dD/d\phi$ and optimize performative risk directly [16, 20], decision-dependent stochastic optimization [11], state-dependent extensions in which the distribution also carries its own dynamics [5, 18], bandit feedback [17], multi-agent decision-dependent games [21], and distributionally robust formulations; see [15] for a survey. Throughout, $(\varepsilon, \beta, \gamma)$ remain abstract constants of an unspecified loss.

The market-making literature supplies exactly the missing structure: closed-form optimal quotes under exponential fill intensities [1, 14], multi-asset dimensionality reduction [4], adverse selection and price reading [3], the general stochastic-control toolkit [6], and the empirical anatomy of flow–price interaction [7]. But these models treat the flow response to quoting as a fixed model input;

the learning loop that re-estimates the market from its own induced data is outside their scope.

To our knowledge no prior work computes the performative constants from market structure, which is precisely what is needed to make the stability theorem operational.

1.3 Proposed Approach

REFLEX (*Reflexive Equilibrium Fixed-point Learning*) bridges the two: it realizes performative prediction *inside* a structural OTC market-making model in the Guéant–Lehalle–Fernández-Tapia (GLFT) family, derives the loop constants from microstructure primitives, and then verifies every closed form against a learned simulator loop. Six results, each derived and then falsified-or-confirmed by experiment:

- **R1** an analytic stability boundary: γ, β, and ε in closed form from fill-curve curvature, P&L scale, and the toxic-flow slope, giving the modulus $m = \varepsilon\beta/\gamma$ *a priori*;
- **R2** a closed-form performative-gradient (PerfGD) correction and its *estimated*, structurally anchored counterpart – the learned loop that remains stable beyond the RRM boundary;
- **R3** a multi-dealer boundary $\varepsilon < \gamma/(N_{\text{eff}}\beta)$: competition destabilizes the market a factor N_{eff} before any single dealer would;
- **R4** a finite-sample robust boundary with an $O(1/\sqrt{n})$ ambiguity radius bought by common random numbers (CRN), issuing stable/unstable/*undecided* certificates;
- **R5** factor-model scaling: the $d\times d$ modulus matrix and its spectral boundary $\rho(M) < 1$ at $O(dk^2)$ cost;
- **R6** lazy deployment: the K-step retraining map $\mu(K) = -m + c^K(1+m)$, with deadbeat and maximal-stability cadences in closed form.

The system side contributes an *un-blinded* learned market-response operator whose training identifies $dD/d\phi$; three independent instruments for ε (a CRN best-response probe, an entropic optimal-transport estimate [8, 12, 23], and a fitted informed-flow curve [7]); calibration of the simulator to 36 years of public market data [10, 13]; and a verification layer of 66 numerical proof certificates plus Lean 4 [9, 24] formal skeletons. Honesty is a design constraint: negative results and proxy-level data provenance are reported as such throughout.

1.4 Results Overview

All 66 certificates pass on raw and real-unit calibrated configurations (worst residual per family 2.2×10^{-16}–6.8×10^{-2}, each within its stated tolerance). On real data, the stability headroom $\varepsilon^* = \gamma/\beta$ collapses ~4.4× (investment grade, IG) and ~4.3× (high yield, HY) from calm to crisis, with HY more than 10× below IG in every regime. Predict-then-verify: the measured modulus tracks the closed form within 8% in the contracting regime, and the measured boundary crossing ($f^* \approx 3.17$) is reconciled with the a-priori prediction (4.70) by an independently measured liquidity-inflation correction ($\rho \approx 2.3$). A genuine two/three-dealer market amplifies the measured modulus 1.74× / 3.16× against predicted $N_{\text{eff}} = 2$ / 3. In a regime where blind RRM collapses (final $h = 0.16$), the structurally anchored PerfGD loop converges to the realized performative optimum ($h = 3.19$, within 0.7% of its own estimate), while a free-form learned correction fails. The lazy-deployment law lands both parameter-free predictions: the sign flip at K_{db} and the stability-window exit at $K_{\max}$.

2 The Market Model and the Performative Loop

One deployment is one period of quoting by a dealer posting half-spread h (skew zero for the scalar theory; the multi-bond lift is R5). With ρ the latent liquidity ratio, benign notional follows the GLFT exponential fill curve and informed notional a spread-gated channel:

$$U(h) = A\rho\, e^{-kh}, \qquad \tau(h) = \rho\, \bar{g}\big(I_b + \alpha f I\, e^{-c_t h}\big), \tag{1}$$

where A, k are the arrival rate and demand elasticity, I_b, I the base and feedback informed intensities, α adversarial informativeness, f the toxicity-feedback gain (our control variable), c_t the toxic spread decay, and $\bar{g} \in (0,1)$ the mean of the informed traders' tanh signal gate. Informed flow trades with the signal, so each unit costs the dealer an adverse-selection severity $\psi > 0$ (a state constant from the same Gaussian gate integrals). The expected one-deployment objective, with P&L scale P, quoting-cost weight w, anchor h_{ref}, and inventory-variance curvature $\lambda_q \ge 0$, is

$$J(h;T) = P\big[\, hA\rho e^{-kh} + hT - \psi T - w(h-h_{\text{ref}})^2 - \lambda_q\,\big]. \tag{2}$$

The structural content of "the learner is blind" is the frozen toxic level T: within a deployment the fitted model conditions on the deployed regime, so the dealer best-responds to $T = \tau(h_{\text{dep}})$ while only the benign channel varies with its candidate h. Retraining is then the cobweb

$$h_{t+1} = \text{BR}(h_t) := \arg\max_h J\big(h;\tau(h_t)\big), \qquad m := \big|\text{BR}'(h^*)\big|, \tag{3}$$

which contracts to the performatively stable spread h_{SP} iff $m < 1$. The simulator realizes (1)–(3) with multi-bond flow, an informed-flow saturation cap, and a liquidity field that *inflates with realized flow* – endogenous channels that the frozen closed forms deliberately omit and that the measurements below make visible.

3 Closed-Form Stability Theory

Full derivations (with proofs, regularity conditions, and the saturated corollaries) are documents D1–D6 in the repository (see the *Reproducibility* statement); we state the objects the experiments test.

R1 (analytic boundary). Differentiating (2), the three Perdomo constants are closed forms in the configuration:

$$\gamma = P\big[2w + A\rho k e^{-kh}(2-kh) + \lambda_q\big], \qquad \beta = P,$$
$$\varepsilon(h) = \rho\,\bar{g}\,\alpha f I c_t\, e^{-c_t h}, \tag{4}$$

i.e. curvature is fill-curve curvature plus quoting-cost stiffness, smoothness is the P&L scale, and ε is the toxic-flow slope $|d\tau/dh|$. The implicit-function theorem gives $\text{BR}' = -\varepsilon\beta/\gamma$ (an oscillatory cobweb), hence

$$m(h^*) = \frac{\varepsilon\beta}{\gamma} = \frac{\rho\bar{g}\alpha f I c_t\, e^{-c_t h^*}}{2w + A\rho k\, e^{-kh^*}(2-kh^*) + \lambda_q}, \tag{5}$$

with P cancelling and h^* the self-consistent fixed point of (3); the loop is stable iff $\varepsilon < \gamma/\beta$, i.e. $m < 1$. Everything is evaluable before any loop is run: the boundary is a *prediction*. Two structural corollaries matter for honest measurement. First, adverse selection (ψT) is constant in h within a deployment, so it sets the profit level (the echo-chamber gap of R2) but drops out of the boundary. Second, at the self-consistent fixed point the modulus *saturates*: widening spreads decay $\varepsilon(h^*)$ exponentially, so at default-like constants the fixed-point m never crosses 1 (defensive widening). Measured instability is therefore a statement about the retraining map at the *operating* spread, and we report it as such.

R2 (PerfGD correction). The performative objective is $\Phi(h) = J(h; \tau(h))$. Un-blinding adds one closed-form term to the blind gradient $G(h) = \partial_1 J(h; \tau(h))$:

$$\Phi'(h) = G(h) + \Delta(h), \qquad \Delta(h) = -\beta\,(h - \psi)\,\varepsilon(h), \tag{6}$$

where $\Delta = \partial_T J \cdot (d\tau/dh)$ is the distribution-response term, supplied entirely by the R1 closed forms – the corrected ascent costs one scalar more per step than RRM [16]. Its convergence is governed not by the cobweb modulus but by the objective curvature at the performative optimum h_{PO},

$$\gamma_{\mathrm{PO}} = \gamma + \beta\varepsilon\,(2 + c_t\psi - c_t h_{\mathrm{PO}}), \tag{7}$$

which stays positive where $m > 1$: the corrected loop converges beyond the RRM boundary. The stable-versus-optimal separation ("echo-chamber gap" [20]) is $h_{\mathrm{SP}} - h_{\mathrm{PO}} = \beta\varepsilon(h_{\mathrm{SP}} - \psi)/(\gamma + \beta\varepsilon) = O(\varepsilon)$ in decision space and $O(\varepsilon^2)$ in value.

R3 (multi-dealer systemic risk). N dealers share one informed pool with toxic spillover $\kappa \in [0, 1]$ (flow displaced by one dealer's widening lands on the others). The joint best-response Jacobian is a rank-one common-mode coupling: its differential eigenvalues carry $(1-\kappa)m_1$ and its common mode $-m_1 N_{\mathrm{eff}}$ with $N_{\mathrm{eff}} = 1 + \kappa(N-1)$, so the joint loop is stable iff

$$\varepsilon < \frac{\gamma}{N_{\mathrm{eff}}\,\beta}, \qquad N_c = 1/m_1 \tag{8}$$

is the critical dealer count at which a market of individually stable dealers ($m_1 < 1$) turns systemically unstable. Competition manufactures a synchronized cobweb – fragility is a market property, not a desk property.

R4 (robust boundary). With n probe estimates of ε (sample mean $\hat{\varepsilon}_n$, std s), the certificate

$$\hat{\varepsilon}_n + \delta_n < \frac{\gamma}{\beta}, \qquad \delta_n = \frac{z_{1-a}\,s}{\sqrt{n}}, \tag{9}$$

declares *stable*; $\hat{\varepsilon}_n - \delta_n > \gamma/\beta$ declares *unstable*; otherwise *undecided*. The parametric $O(1/\sqrt{n})$ radius is bought by the CRN pairing of the probe (a naive finite difference of independent runs achieves only $O(n^{-1/3})$ after bias–variance balancing), and pinning a crossing needs $n_{\mathrm{req}} = O(\Delta^{-2})$ in the distance Δ to the boundary – statistical uncertainty is separated from structural uncertainty by construction. A distribution-free quantile calibration of δ_n (new here) shows $z\,s$ is conservative except under contamination (Sec. 5.7).

R5 (factor scaling). For d bonds, stability is governed by the modulus matrix $M = \beta\,\Gamma^{-1}E$ with $E = \mathrm{diag}(\varepsilon_i)$ and Γ the cross-bond curvature; the loop contracts iff $\rho(M) < 1$. A k-factor covariance makes Γ diagonal-plus-low-rank, so $\rho(M)$ costs $O(dk^2)$ by Woodbury, with truncation error linear in the residual factor variance $\lambda_{k+1}(C)$: the same structure that creates dimensionality defuses it.

R6 (lazy deployment). Suppose each deployment takes only K inner gradient steps on the frozen-T objective (step η_h, contraction $c = 1 - \eta_h\gamma$), warm-started from the deployed spread, instead of the exact best response. The outer map slope becomes

$$\mu(K) = -m + c^K(1 + m), \tag{10}$$

a one-parameter interpolation from inertia ($\mu(0) = 1$) to the exact cobweb ($\mu \to -m$). Consequences, all closed-form: a *deadbeat* cadence $K_{\mathrm{db}} = \ln\big(m/(1+m)\big)/\ln c$ at which $\mu = 0$ (one-shot convergence); for $m > 1$ a stability window $K \le K_{\max} = \ln\big((m-1)/(m+1)\big)/\ln c$ – laziness keeps an RRM-unstable market stable, the quantitative form of the greedy-vs-lazy gap in [19, 22]; and an observer who fits the plain boundary to a lazy loop reads a two-branch *effective* curvature $\gamma_{\mathrm{eff}}(K) = \gamma\,m/|\mu(K)|$, under-estimating stiffness below the equal-modulus cadence and over-estimating it above.

4 System and Measurement Methodology

4.1 Learned Operator and the Four Loop Modes

The simulator is a multi-bond OTC market with benign and informed clients, inventory, an informed-flow saturation cap, and a latent liquidity field inflated by realized flow. The learned market-response operator T_θ (MLP heads over per-bond features) predicts next-deployment flow distributions conditioned on a per-deployment *policy summary*, and is fit over a sliding window of past deployments; with window ≥ 2 the derivative of its prediction w.r.t. the summary – the learned $dD/d\phi$ – is identified by automatic differentiation, whereas the single-window fit is structurally blind (the RRM baseline). Four retraining modes close the loop: **(i)** blind RRM; **(ii)** *PerfGD-analytic*, using the closed-form Δ of (6) as a surrogate gradient; **(iii)** *PerfGD-learned*, consuming the free-form learned $dD/d\phi$ (reported as a negative result); and **(iv)** *PerfGD-structural*, which never asks the network for a derivative: it fits the theory's own response families

$$\hat{\tau}(h) = \hat{C}_0 + \hat{C}_1 e^{-\hat{c}h}, \qquad \hat{u}(h) = \hat{A}_u e^{-\hat{k}_u h}, \tag{11}$$

plus the realized severity $\hat{\psi}$, to the loop's *own* deployment history (per-bond, per-step samples over a 12-deployment window; a small quote jitter provides within-deployment identification that the echo chamber cannot destroy), then ascends $\hat{\Phi}' = \hat{G} + \hat{\Delta}$ with step $1/\gamma_{\hat{\mathrm{PO}}}$ under a trust region (35% relative step cap) and an anti-echo freeze that refuses to steer on unidentified fits. Each safeguard encodes a measured failure: exponential response fits on a narrow spread window are line-degenerate and extrapolate arbitrarily badly, and the unconstrained early ascent rails against its spread cap.

4.2 Three Instruments for ε and the Probe Protocol

(a) *CRN best-response probe*: paired deployments at $h_{\mathrm{ref}} \pm \delta$ under common random numbers; the finite-difference of best responses estimates m (and $\varepsilon = \hat{m}\gamma/\beta$), and a signed K-step variant measures $\mu(K)$ for R6. (b) *Optimal transport*: $\hat{\varepsilon} = W_1\big(D(h+\delta), D(h-\delta)\big)/2\delta$ – the exact Wasserstein sensitivity of performative prediction – via debiased log-domain Sinkhorn divergences [8, 12], with the blur set *scale-relatively* (0.02× pooled std, tuned against the exact 1-D quantile W_1; Sec. 5.7). (c) *Fitted informed-flow curve*: bin realized toxic notional against deployed spread, fit (1), differentiate [7]. Protocol facts that a pre-run audit showed matter at first order: probe at the operating spread (not the analytic fixed point); collection jitter 0.05 (at 0.2 the BR probe inflates ∼3×); medians and interquartile ranges

(IQR) over 8 seeds with the R4 robust bands at every grid point. We sweep the gain f, not adversariality α: high α drives the dealer to wide spreads where $e^{-c_t h}$ has decayed, so the α-response is non-monotone (measured hump $0.08 \to 1.83 \to 0.67$) – a confound, and the quantitative case for f as the control variable.

4.3 Real-Data Calibration and Provenance

The simulator is calibrated per (rating × volatility regime) on a public, verified panel: ~36 years of daily and ~70 years of monthly series (CBOE VIX as the σ proxy and regime classifier; WTI; the Fed H.15 10-year; Shiller equity data; gold/CPI), TRACE-derived corporate-bond factors of [10] (the liquidity risk factor is the primary ε proxy), and monthly returns for 212 real-CUSIP bonds. Half-spread proxies combine VIX-implied spreads with the illiquidity add-on of [2, 13], and $\lambda(h) = Ae^{-kh}$ intensities are fit per cell by maximum likelihood. *Provenance, stated plainly:* this is not trade-level TRACE. Dealer-side prints, per-dealer inventories, and per-bond (A, k) would require TRACE Enhanced; only (A, k, σ, h) are data-identified here, the toxic channel is structurally scaled by documented ratios, and the crisis-regime intensity fit is degenerate ($k = 0$, $n = 74$ days) and flagged wherever it binds. Regime *ordering* of the boundary is data-driven; absolute critical gains are not. Calibrated configurations run in per-$100-par units, and every probe width, tolerance, and OT blur is scale-relative by convention.

4.4 Verification Layer

Every load-bearing identity, inequality, and dynamical claim of R1–R6 is re-derived numerically by 66 *proof certificates* run on both the raw and the calibrated real-unit configurations: central differences of the *composed* best-response map against the assembled $-\varepsilon\beta/\gamma$ (no shared code path), eigensolves against spectral formulas, Monte-Carlo rate fits against claimed exponents, and the actual dynamics run rather than assumed. Building the layer caught real errors before the paper-grade run: a convention mismatch in which the 1-D frozen-gradient helpers omit the inventory curvature (slopes governed by $\gamma - P\lambda_q$) while the full pipeline uses the λ_q-inclusive γ; and two falsified first drafts (an "always-stiffer" γ_{eff} claim, corrected to the two-branch law of R6; and a mis-signed heavy-tail prediction for the R4 radius). The logical skeletons of all six results are additionally formalized in Lean 4 against mathlib [9, 24]; these are reviewed statements whose compilation is pending a toolchain, so the numerical certificates are the verification of record. The full suite (152 tests, 11 experiments) runs CPU-only in ~25 minutes, deterministic from (configuration, seed).

5 Results

5.1 Machine-Checked Identities

Table 1 summarizes the certificates: 66/66 pass, with worst residuals far inside tolerance. The one excluded cell is deliberate: the R2 beyond-boundary *demo dynamics* certificate pins absolute raw-unit constants, and imposing those on per-$100-par calibrated units is exactly the class of unit bug the conventions forbid – it excludes itself and the calibrated pass proves the identities in real units.

Table 1: Numerical proof certificates (raw + calibrated real-unit configurations). Each family's worst residual sits inside its stated tolerance.

Certificate family	raw	calib.	worst resid.	tol.
R1 analytic boundary	5/5	5/5	6.8×10^{-2}	8×10^{-2}
R2 PerfGD identities	4/4	4/4	1.3×10^{-6}	2×10^{-2}
R2 PerfGD demo dynamics	4/4	–	1.4×10^{-11}	1×10^{-3}
R3 multi-dealer	6/6	6/6	2.2×10^{-16}	1×10^{-9}
R4 robust boundary	7/7	7/7	3.2×10^{-2}	1×10^{-1}
R5 factor scaling	3/3	3/3	0.0	1×10^{-9}
R6 lazy deployment	6/6	6/6	1.1×10^{-16}	1×10^{-12}

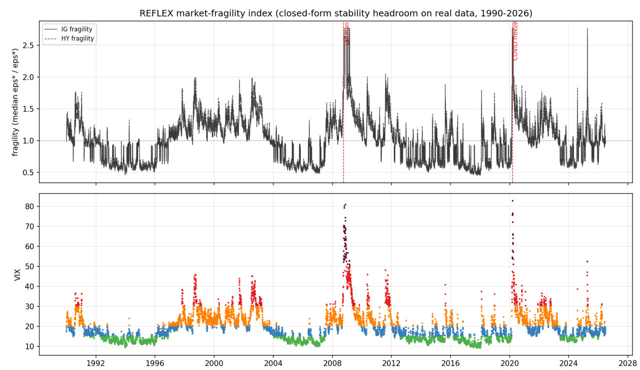

Figure 1: The daily market-fragility index, 1990–2026: the R1 closed forms evaluated on real data. Top: fragility(t) = median$(\varepsilon^*)/\varepsilon^*(t)$ **for IG (solid) and HY (dashed), with the Lehman and COVID-freeze dates marked. Bottom: the VIX spine, colored by volatility regime. The crisis plateau reflects the degenerate crisis-cell fit and is flagged, not hidden.**

5.2 A Real-Data Fragility Index (1990–2026)

Because R1 is a closed form of observables, it can be evaluated on every trading day of the panel. Figure 1 plots the resulting index, fragility(t) = median$(\varepsilon^*)/\varepsilon^*(t)$, and Table 2 its per-regime summary. Three facts survive all caveats: the stability headroom $\varepsilon^* = \gamma/\beta$ collapses ~4.4× (IG) and ~4.3× (HY) from calm to crisis; HY sits more than 10× below IG in *every* regime (a level, not a crisis, effect); and the modulus at *observed* spreads falls into crisis (IG $0.85 \to 0.14$) – dealers widen faster than the toxic channel steepens, i.e. defensive widening is visible in real data. The index saturates at a crisis plateau (the degenerate crisis fit): the GFC and the March-2020 COVID freeze are flagged at the same ceiling, and intra-crisis ranking is explicitly not identified. Per-cell a-priori boundaries span $\varepsilon^* = 465 \to 2.2$ from IG-calm to HY-crisis while the fixed-point modulus stays regime-invariant (≈0.066) by construction – the regime story lives entirely in the headroom.

5.3 Predict-Then-Verify: the Boundary Crossing

Figure 2 is the headline falsification test: sweep the gain f (7 values × 8 seeds), overlay the a-priori curve (5), and measure the modulus with the CRN probe. In the contracting regime the agreement is

Table 2: Per-regime stability headroom $\varepsilon^* = \gamma/\beta$ and modulus at observed spreads, evaluated on the 1990–2026 panel.

regime	$\varepsilon^*_{\mathrm{IG}}$	$\varepsilon^*_{\mathrm{HY}}$	$m_{\mathrm{IG}}(h_{\mathrm{obs}})$	$m_{\mathrm{HY}}(h_{\mathrm{obs}})$
calm	1207.7	93.5	0.847	0.592
normal	739.2	61.3	0.870	0.563
elevated	594.4	45.9	0.520	0.367
stress	477.3	35.6	0.223	0.163
crisis	275.9	21.8	0.139	0.091

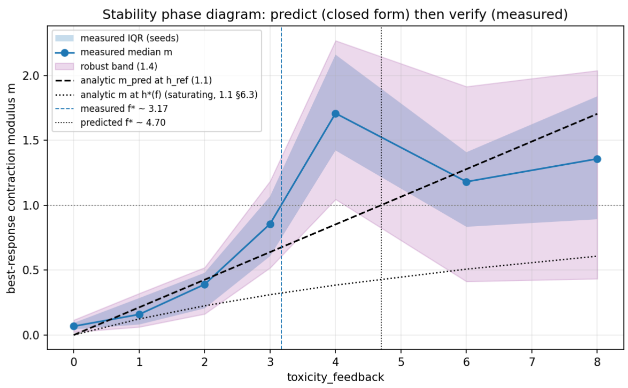

Figure 2: Predict-then-verify phase diagram over the feedback gain f: measured modulus (median, IQR band, and R4 robust band over 8 seeds) against the a-priori closed form at the probe spread and the saturating fixed-point curve, with the measured ($f^* \approx 3.17$) and predicted (4.70) crossings marked.

quantitative: measured median 0.390 versus predicted 0.426 at $f = 2$ (8%). The measured crossing sits at $f^* \approx 3.17$ against the a-priori 4.70 – and the gap is not free: the triangulation (Table 3) independently measures the deployment's own flow inflating the liquidity field to $\rho \approx 2.3$, and the closed form evaluated at that *realized* state predicts $f^* \approx 2.8$–3.0, bracketing the measurement. The R4 certificates grade the grid exactly as the seed bands warrant: stable through $f = 2$, unstable at $f = 4$, undecided where beyond-boundary readings scatter (at $f = 8$ the eight seeds span 0.03–2.05; past the boundary the probe is a finite-difference diagnostic, not a local slope). The fixed-point curve itself saturates at 0.61 – the measured instability is a property of the retraining map at the operating spread, exactly as R1 predicts.

The two distribution-space instruments agree with each other within 14% and sit 2.3–2.7× above the realized-state closed form: the residual is the state-feedback channel that any frozen-state closed form omits by construction, so the analytic boundary is a *lower anchor*, not an unbiased point prediction – the operationally safe direction for a stability certificate.

5.4 Competition Amplifies Instability (R3)

On a *genuine* N-dealer market sharing one informed pool (not an analytic shortcut; the environment reduces bit-for-bit to the single-dealer market at $N = 1$), CRN joint-mode probes measure

Table 3: Three-way ε triangulation at the operating spread, against the closed form at the a-priori (A2) and at the realized deployment state ($\rho = 2.32$, $|g| = 0.446$).

instrument	$\hat{\varepsilon}$	vs. realized
closed form, a-priori state ($\rho = 1$)	0.764	0.44×
closed form, **realized state**	1.726	1.00
CRN best-response slope	0.508	0.29×
Sinkhorn/W_1	4.578	2.65×
Fitted informed-flow curve	4.011	2.32×

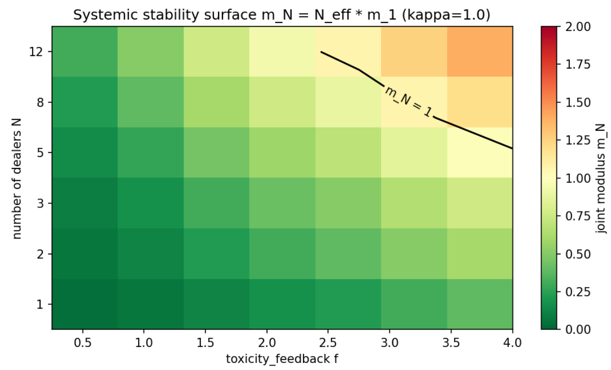

Figure 3: The R3 systemic surface: joint modulus $m_N = N_{\mathrm{eff}}\, m_1$ over the (f, N) grid at full spillover ($\kappa = 1$), with the $m_N = 1$ boundary contour. Gains that are comfortably stable for a single dealer cross the systemic boundary as the dealer count grows; the probes of Table 4 measure the first rows of this surface on the simulated market.

Table 4: Measured common-mode retraining slope on the shared-pool market versus the R3 prediction $N_{\mathrm{eff}} m_1$ (full spillover, $\kappa = 1$).

N	measured	predicted	differential mode
1	0.786	0.786	–
2	1.369	1.571	0.0034
3	2.480	2.357	0.0032

the common-mode slope against the predicted $N_{\mathrm{eff}} m_1$ (Table 4): amplification 1.74× (N=2) and 3.16× (N=3) against predicted 2 and 3 – within 13% and 5% – while the differential mode is dead at full spillover ($(1-\kappa)m_1 = 0$), i.e. the instability is purely the synchronized systemic channel. The analytic surface (Figure 3) puts the critical dealer count at $N_c \approx 7.9$ for the default single-dealer modulus, and the simulated three-dealer cobweb already oscillates against its cap. The governance reading is direct: each desk can satisfy its own stability check while the market as a whole violates (8); fragility must be certified at the market level.

Table 5: Loop outcomes in the RRM-unstable demo regime (10 deployments, common seed; convergence verified against independent structural fits).

mode	final h	outcome
blind RRM	0.16	echo-chamber collapse
PerfGD-analytic	0.36	no settle (operator gradient)
PerfGD-learned	0.66	no settle (negative result)
PerfGD-structural	**3.19**	**converges to realized optimum**

5.5 Closing the Loop-Level Gap: Structural PerfGD (R2)

The demo regime is genuinely RRM-unstable ($f = 6$, slow toxic decay; cobweb modulus 1.21): the blind cobweb provably diverges while the corrected 1-D ascent converges – in closed form. The question the learned system must answer is whether a *learned* loop can realize the correction. Figure 4 and Table 5 give the answer from a common seed. Blind RRM collapses into the echo chamber (final half-spread 0.16). PerfGD-analytic and the free-form PerfGD-learned mode both fail to settle: each consumes the operator's implied objective gradient, and the run's seam diagnostics show the free-form learned toxic slope starting right-signed (-0.84 versus analytic -1.79) and then flipping sign and hovering near zero – capacity is not the problem, identification off the deployed regime is. The structural mode never asks the network for a derivative: fitting (11) to its own deployment history, it climbs under the trust region and *converges to the realized performative optimum* ($h = 3.193$, late steps $0.35 \to 0.02$), within 0.7% of its own running estimate $\hat{h}_{\mathrm{PO}} = 3.172$, its fitted slope tracking the analytic shape (-1.33 vs. -2.00 at the final iterate) while strengthening as the fit window widens.

Benchmark honesty. The frozen-reference closed-form optimum ($h_{\mathrm{PO}} = 1.641$) is *not* the target: in this high-intensity regime the realized market differs from the frozen closed forms at first order through channels they omit by construction – informed-flow saturation, the $\rho \approx 2.3$ liquidity inflation of Sec. 5.3, and severity drift. The loop is therefore verified against *independent structural fits* on fresh controlled deployments spanning the operating range: the settle point (i) zeroes their corrected gradient (residual < 30% of its start value), (ii) sits strictly inside their blind stable point ($< 0.85 \times \hat{h}_{\mathrm{SP}}$ – the realized echo-chamber gap, closed), and (iii) brackets their independent optimum estimate. The scoped claim: un-blinding is proven in closed form, and the structurally anchored learned loop realizes it at loop level against the realized-market benchmark; the free-form learned correction remains a documented negative result. Anchoring, not capacity, closes the gap.

5.6 Lazy Deployment Stabilizes an Unstable Market (R6)

At a beyond-boundary configuration ($f = 5$; exact-BR anchor $\hat{m} = 1.205 > 1$ measured on the same seeds), the signed CRN K-step probe traces $\mu(K)$ for $K \in \{1, 2, 3, 5, 8, 12, 20\}$ (Figure 5). A one-parameter fit of (10) gives the inner contraction $c = 0.661$, and both parameter-free predictions land: the deadbeat sign flip at $K_{\mathrm{db}} = 1.46$ (measured medians $+0.021$ at K=1, -0.064 at K=2) and the stability-window exit at $K_{\max} = 5.75$ (measured $|\mu|$: 0.695 at K=5, inside; 1.541 at K=8, outside). Laziness measurably keeps this RRM-unstable market stable for $K \lesssim 6$ and inherits the cobweb's divergence beyond – retraining *cadence* is a stability control. The measured $\gamma_{\mathrm{eff}}/\gamma$ falls monotonically through 1 between $K = 5$ and $K = 8$, the stiff branch of the two-branch law. Beyond-boundary per-seed scatter is large by construction (medians carry the result); the clean tight-fit demonstration lives in the contracting regime (measured +0.78/+0.70/+0.37 vs. fitted +0.88/+0.68/+0.36 at $K = 1/3/8$) and is locked in the test suite.

5.7 Scaling and Estimator Tuning

Factor scaling (R5). With per-bond volatilities dispersed at the data-calibrated cross-sectional coefficient of variation, $\rho(M)$ is flat at $\approx$0.50 from $d = 8$ to 128 bonds (Figure 6) and pinned to the worst scalar modulus – on this calibration, correlation does not manufacture cross-sectional instability; the fragile mode is idiosyncratic. The Woodbury path computes $d = 128$ in 0.12 s and matches the dense eigensolve to machine precision; the truncation bound holds with 3–4 orders of magnitude of slack. *Sinkhorn blur (R1 instrument).* Against the exact 1-D quantile W_1, the debiased-divergence bias is U-shaped in the scale-relative blur at a fixed iteration budget (log-domain under-convergence below, entropic over-blur above), with its minimum at 0.02× std (6.1% ground-truth bias; 1.2% on the config's CRN samples) – baked in as the default and, being scale-relative, transferring to real-unit configurations unchanged. *Robust radius (R4).* Across normal, heavy-tailed, and skewed estimate distributions the $z\,s$ radius over-covers (0.99–1.00 at 95% nominal); only contamination (railed-probe patterns) degrades both radii (0.938) – the honest n_{req} limit. On the actual CRN probe estimates the quantile/$z\,s$ multiplier is 0.50, so $z\,s$ is binding and adequately calibrated; the quantile-guarded radius is kept for suspected contamination.

6 Conclusion

REFLEX turns the performative-prediction stability theorem from an abstraction into an instrument: $(\varepsilon, \beta, \gamma)$ computed from microstructure primitives, six closed-form extensions (correction, competition, finite samples, dimension, cadence), each verified predict-then-verify against a learned loop, evaluated as a daily fragility index on 36 years of real market data, and machine-checked end to end. Two findings matter beyond this model. First, competition among retraining dealers amplifies instability by the effective dealer count – a market-level, not desk-level, certification problem for algorithmic liquidity provision. Second, a learned corrected loop stabilizes where blind retraining collapses *only when its response model is anchored to market structure*: the free-form alternative fails with the same data and more capacity. Structure is not a nicety here; it is the difference between a loop that converges and one that does not.

7 Limitations and Future Work

Data. The calibration is proxy-level, not trade-level TRACE: VIX-implied spreads, no per-dealer inventories, a structurally scaled toxic channel, and a degenerate crisis cell ($k = 0$) – regime ordering

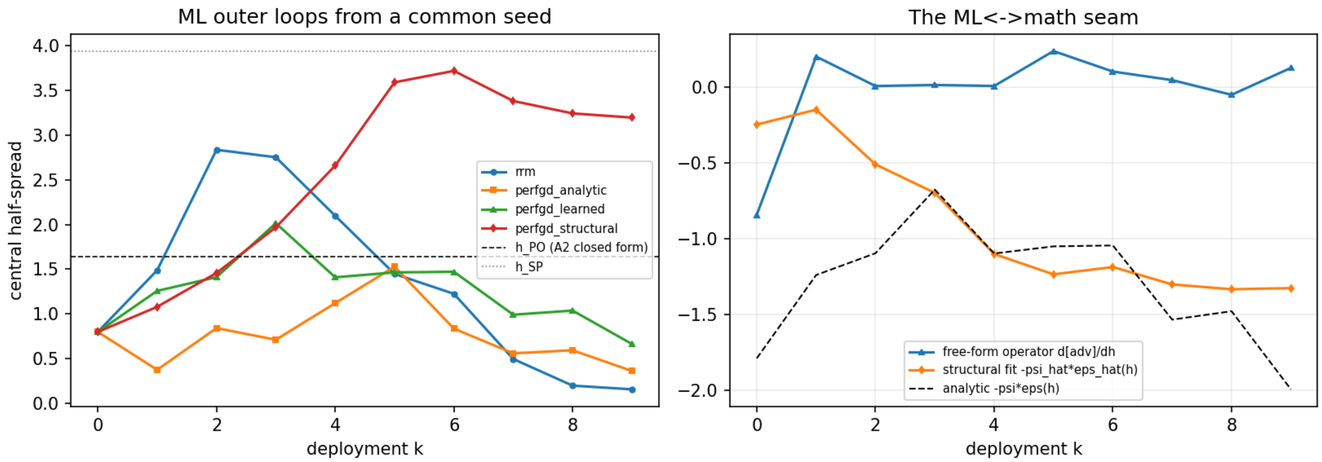

Figure 4: Left: half-spread trajectories of the four learned retraining loops from a common seed in the RRM-unstable regime (the frozen-reference h_{PO} and h_{SP} lines are context, not the benchmark; Sec. 5.5). Blind RRM collapses; the structurally anchored loop converges to the realized performative optimum. Right: the three-way ML↔theory seam – the free-form learned, structurally fitted, and analytic toxic slopes logged at every deployment. The free-form slope flips sign and hovers near zero; the structural fit tracks the analytic shape.

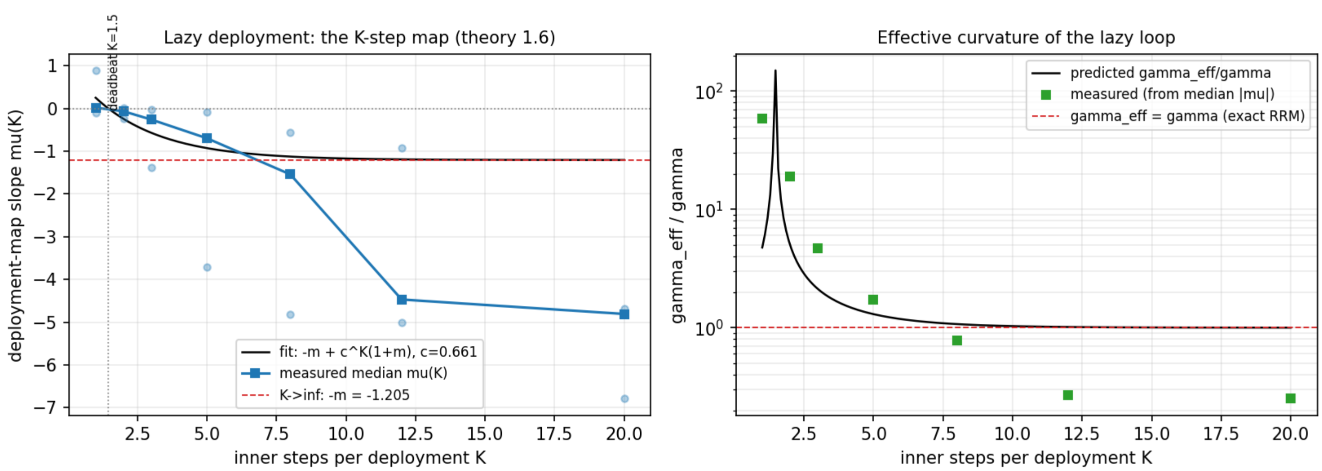

Figure 5: Lazy deployment at a beyond-boundary configuration ($\hat{m} = 1.205$). Left: the measured signed K-step map (medians over seeds; faint dots are per-seed readings) against the one-parameter fit $\mu(K) = -m + c^K(1 + m)$ with $c = 0.661$, its $K \to \infty$ asymptote $-m$, and the predicted deadbeat cadence. Right: the implied effective curvature $\gamma_{\mathrm{eff}}(K)/\gamma$, predicted versus measured – the stiff branch decaying through the exact-RRM line between $K = 5$ and $K = 8$.

is data-driven, absolute critical gains are not. TRACE Enhanced calibration is the direct upgrade path. *Measurement.* At default-like constants the self-consistent fixed point never destabilizes (defensive widening), so measured crossings are statements about the retraining map at the operating spread; beyond the boundary, probe readings are finite-difference diagnostics with seed-level bifurcation (medians and robust bands carry the results); measured moduli are protocol-dependent and comparable within, not across, protocols. *The structural loop.* Its optimum is the *realized* one – benchmarked against independent structural fits, never the frozen-reference closed form, with the gap channel-attributed (saturation, liquidity inflation, severity drift); its convergence is to the noise ball of an estimated gradient, and its early iterations lean on the trust region. The free-form learned correction remains a negative result by design. *Verification.* The Lean 4 skeletons are reviewed but not yet compiled; the 66 numerical certificates are the verification of record. Future work: trade-level calibration, inventory-state loop

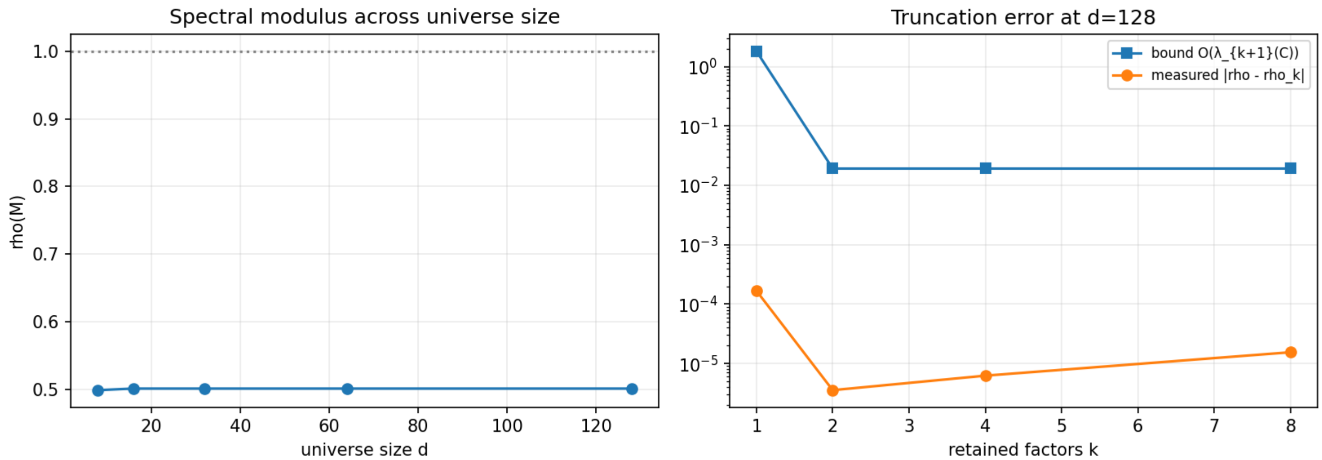

Figure 6: Factor-model scaling (R5) with data-calibrated per-bond dispersion. Left: $\rho(M)$ is flat in universe size from $d = 8$ to 128 bonds and pinned to the worst scalar modulus, far below the $\rho = 1$ boundary (dotted). Right: the truncation bound $O(\lambda_{k+1}(C))$ versus the measured error $|\rho - \rho_k|$ at $d = 128$ – three to four orders of magnitude of slack at every retained-factor count k.

dynamics (state-dependent performativity), richer policy classes, and compiling the formal layer.

Reproducibility

The full framework – the six derivation documents D1–D6 (each with a compilable LaTeX twin), the simulator, operator, all four loop modes, the estimator suite, the calibration pipeline with its data catalogue and rejected-sources log, the 66-certificate verification layer, the Lean sources, every configuration and seed, and the complete illustrated reports of both paper-grade runs – is openly available.[1] All results in this paper are regenerated by one command (`run_all -profile full`, ~25 CPU-minutes, deterministic from configuration and seed).

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